CS 395T Homework 4

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Problem 1

(i)

Proof. By the equivalent characterization of total variation (Remark 1, Lecture 14),

$$D_{\mathrm{TV}}(\mathcal{T}\pi, \pi^\star) = \sup_{A \subseteq \mathbb{R}^d} \left(\int_A (\mathcal{T}\pi)(\mathbf{y})\, \mathrm{d}\mathbf{y} - \int_A \pi^\star(\mathbf{y})\, \mathrm{d}\mathbf{y} \right).$$

Fix a measurable set $A \subseteq \mathbb{R}^d$. Expanding $(\mathcal{T}\pi)(\mathbf{y})$ and using stationarity of $\pi^\star$:

$$\begin{aligned}
\int_A (\mathcal{T}\pi)(\mathbf{y})\, \mathrm{d}\mathbf{y} - \int_A \pi^\star(\mathbf{y})\, \mathrm{d}\mathbf{y} &= \int_A \int_{\mathbb{R}^d} \mathcal{T}_{\mathbf{x}}(\mathbf{y})\, \pi(\mathbf{x})\, \mathrm{d}\mathbf{x}\, \mathrm{d}\mathbf{y} - \int_A \int_{\mathbb{R}^d} \mathcal{T}_{\mathbf{x}}(\mathbf{y})\, \pi^\star(\mathbf{x})\, \mathrm{d}\mathbf{x}\, \mathrm{d}\mathbf{y} \\
&= \int_{\mathbb{R}^d} \left(\int_A \mathcal{T}_{\mathbf{x}}(\mathbf{y})\, \mathrm{d}\mathbf{y} \right) \big(\pi(\mathbf{x}) - \pi^\star(\mathbf{x})\big)\, \mathrm{d}\mathbf{x},
\end{aligned}$$

where the interchange of integrals is justified by Fubini's theorem (the integrands are non-negative densities, hence measurable, and the product measures are σ-finite).

Since $\mathcal{T}_{\mathbf{x}}$ is a probability distribution for each $\mathbf{x}$, we have $\int_A \mathcal{T}_{\mathbf{x}}(\mathbf{y})\, \mathrm{d}\mathbf{y} \leq 1$. Define $g(\mathbf{x}) := \int_A \mathcal{T}_{\mathbf{x}}(\mathbf{y})\, \mathrm{d}\mathbf{y} \in [0, 1]$. Then

$$\begin{aligned}
\int_{\mathbb{R}^d} g(\mathbf{x}) \big(\pi(\mathbf{x}) - \pi^\star(\mathbf{x})\big)\, \mathrm{d}\mathbf{x} &= \int_{\mathbb{R}^d} g(\mathbf{x}) \big(\pi(\mathbf{x}) - \pi^\star(\mathbf{x})\big)^+\, \mathrm{d}\mathbf{x} - \int_{\mathbb{R}^d} g(\mathbf{x}) \big(\pi(\mathbf{x}) - \pi^\star(\mathbf{x})\big)^-\, \mathrm{d}\mathbf{x} \\
&\leq \int_{\mathbb{R}^d} \big(\pi(\mathbf{x}) - \pi^\star(\mathbf{x})\big)^+\, \mathrm{d}\mathbf{x}, \quad \text{(since } g \leq 1 \text{ and } g \geq 0\text{)}
\end{aligned}$$

where $(a)^+ = \max(a, 0)$ and $(a)^- = \max(-a, 0)$.

By the standard identity relating the positive part to TV distance,

$$\int_{\mathbb{R}^d} \big(\pi(\mathbf{x}) - \pi^\star(\mathbf{x})\big)^+\, \mathrm{d}\mathbf{x} = \sup_{B \subseteq \mathbb{R}^d} \left(\int_B \pi(\mathbf{x})\, \mathrm{d}\mathbf{x} - \int_B \pi^\star(\mathbf{x})\, \mathrm{d}\mathbf{x} \right) = D_{\mathrm{TV}}(\pi, \pi^\star),$$

where the first equality holds by taking $B = \{\mathbf{x} : \pi(\mathbf{x}) > \pi^\star(\mathbf{x})\}$, and the second is the equivalent characterization (Remark 1, Lecture 14).

Since the bound $D_{\mathrm{TV}}(\pi, \pi^\star)$ holds for every measurable A, taking the supremum over A gives

$$D_{\mathrm{TV}}(\mathcal{T}\pi, \pi^\star) \leq D_{\mathrm{TV}}(\pi, \pi^\star).$$

□

(ii)

Proof. We show $(\mathcal{T}\pi)(\mathbf{y}) \leq \beta\,\pi^\star(\mathbf{y})$ for all $\mathbf{y} \in \mathbb{R}^d$. Since π is β-warm with respect to $\pi^\star$ (Definition 2, Lecture 15), we have $\pi(\mathbf{x}) \leq \beta\,\pi^\star(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^d$. Then for any $\mathbf{y} \in \mathbb{R}^d$,

$$\begin{aligned}
(\mathcal{T}\pi)(\mathbf{y}) &= \int_{\mathbb{R}^d} \mathcal{T}_{\mathbf{x}}(\mathbf{y})\,\pi(\mathbf{x})\,\mathrm{d}\mathbf{x} \\
&\leq \int_{\mathbb{R}^d} \mathcal{T}_{\mathbf{x}}(\mathbf{y})\,\beta\,\pi^\star(\mathbf{x})\,\mathrm{d}\mathbf{x} \quad (\pi(\mathbf{x}) \leq \beta\,\pi^\star(\mathbf{x}) \text{ and } \mathcal{T}_{\mathbf{x}}(\mathbf{y}) \geq 0) \\
&= \beta \int_{\mathbb{R}^d} \mathcal{T}_{\mathbf{x}}(\mathbf{y})\,\pi^\star(\mathbf{x})\,\mathrm{d}\mathbf{x} \\
&= \beta\,(\mathcal{T}\pi^\star)(\mathbf{y}) \\
&= \beta\,\pi^\star(\mathbf{y}), \quad (\text{stationarity of } \pi^\star)
\end{aligned}$$

so $\mathcal{T}\pi$ is β-warm with respect to $\pi^\star$. □

Problem 2

(i)

Algorithm (Rejection Sampling):

1. Sample $\omega \sim Q$ using one query to $\mathcal{O}$.
2. Query the value oracles for $f(\omega)$ and $g(\omega)$.
3. Accept ω with probability $\frac{f(\omega)}{C \cdot g(\omega)} \in [0, 1]$.
4. If rejected, return to step 1. If accepted, output ω.

The acceptance probability is well-defined since $\frac{f(\omega)}{g(\omega)} \leq C$ implies $\frac{f(\omega)}{C \cdot g(\omega)} \leq 1$, and $f \geq 0$, $g > 0$ ensures $\frac{f(\omega)}{C \cdot g(\omega)} \geq 0$.

Correctness:

Proof. Let $Z_f = \int_\Omega f(\omega)\,\mathrm{d}\omega$ and $Z_g = \int_\Omega g(\omega)\,\mathrm{d}\omega$, so $P(\omega) = \frac{f(\omega)}{Z_f}$ and $Q(\omega) = \frac{g(\omega)}{Z_g}$. The probability that a sample $\omega \sim Q$ is accepted and falls in a measurable set $A \subseteq \Omega$ is:

$$\int_A Q(\omega) \cdot \frac{f(\omega)}{C \cdot g(\omega)}\,\mathrm{d}\omega = \int_A \frac{g(\omega)}{Z_g} \cdot \frac{f(\omega)}{C \cdot g(\omega)}\,\mathrm{d}\omega = \frac{1}{C \cdot Z_g} \int_A f(\omega)\,\mathrm{d}\omega.$$

Conditioning on acceptance, the density of the accepted sample over A is:

$$\frac{\frac{1}{C \cdot Z_g} \int_A f(\omega)\,\mathrm{d}\omega}{\frac{1}{C \cdot Z_g} \int_\Omega f(\omega)\,\mathrm{d}\omega} = \frac{\int_A f(\omega)\,\mathrm{d}\omega}{Z_f},$$

which is precisely $\int_A P(\omega)\,\mathrm{d}\omega$. Hence the accepted sample is distributed according to P. □

Query complexity:

Proof. The acceptance probability in each iteration is:

$$\Pr[\text{accept}] = \int_\Omega Q(\omega) \cdot \frac{f(\omega)}{C \cdot g(\omega)} \, \mathrm{d}\omega = \frac{1}{C \cdot Z_g} \int_\Omega f(\omega) \, \mathrm{d}\omega = \frac{Z_f}{C \cdot Z_g}.$$

The number of iterations until acceptance is geometric with success probability $\frac{Z_f}{C \cdot Z_g}$, so the expected number of iterations is:

$$\frac{C \cdot Z_g}{Z_f} = \frac{C \int_\Omega g(\omega) \, \mathrm{d}\omega}{\int_\Omega f(\omega) \, \mathrm{d}\omega} = N.$$

Each iteration uses one query to $\mathcal{O}$ and one value oracle query to each of f and g, giving at most N queries of each type in expectation. □

(ii)

We apply the rejection sampling algorithm from part (i) with the following choices:

- $\Omega = \mathbb{B}(\mathbf{0}_d, 1)$,
- the unnormalized target density $\tilde{\pi}(\mathbf{x}) = \exp(-f(\mathbf{x}))$ (playing the role of f in part (i)),
- $g(\mathbf{x}) = 1$ for all $\mathbf{x} \in \Omega$, so $Q = \text{Uniform}(\mathbb{B}(\mathbf{0}_d, 1))$.

Computing C: Since f is 1-Lipschitz and $\mathbb{B}(\mathbf{0}_d, 1)$ has diameter at most 2, for all $\mathbf{x} \in \mathbb{B}(\mathbf{0}_d, 1)$:

$$|f(\mathbf{x}) - f(\mathbf{0})| \leq \|\mathbf{x} - \mathbf{0}\|_2 \leq 1,$$

so $f(\mathbf{x}) \geq f(\mathbf{0}) - 1$, which gives:

$$\exp(-f(\mathbf{x})) \leq \exp(-f(\mathbf{0}) + 1) = e \cdot \exp(-f(\mathbf{0})).$$

Since $g(\mathbf{x}) = 1$, we need $\frac{\exp(-f(\mathbf{x}))}{g(\mathbf{x})} \leq C$. Set

$$C = e \cdot \exp(-f(\mathbf{0})),$$

which requires one value oracle query to f (at $\mathbf{0}$).

Algorithm:

1. Query $f(\mathbf{0})$ and set $C = e \cdot \exp(-f(\mathbf{0}))$.
2. Sample $\mathbf{x}$ uniformly from $\mathbb{B}(\mathbf{0}_d, 1)$.
3. Query $f(\mathbf{x})$ and accept with probability

$$\frac{\exp(-f(\mathbf{x}))}{C \cdot g(\mathbf{x})} = \frac{\exp(-f(\mathbf{x}))}{e \cdot \exp(-f(\mathbf{0}))} = \exp(f(\mathbf{0}) - f(\mathbf{x}) - 1).$$

   This lies in $[0, 1]$: the bound $\frac{\exp(-f(\mathbf{x}))}{g(\mathbf{x})} \leq C$ gives $\frac{\exp(-f(\mathbf{x}))}{C \cdot g(\mathbf{x})} \leq 1$, and $\exp(-f(\mathbf{x})) > 0$ gives ≥ 0.

4. If rejected, return to step 2. If accepted, output $\mathbf{x}$.

Correctness follows from part (i): the output is distributed according to $\pi \propto \exp(-f)$ over $\mathbb{B}(\mathbf{0}_d, 1)$.

Query complexity: By part (i), the expected number of iterations is:

$$N = \frac{C \int_{\mathbb{B}(\mathbf{0}_d,1)} g(\mathbf{x})\,\mathrm{d}\mathbf{x}}{\int_{\mathbb{B}(\mathbf{0}_d,1)} \exp(-f(\mathbf{x}))\,\mathrm{d}\mathbf{x}} = \frac{e \cdot \exp(-f(\mathbf{0})) \cdot \mathrm{Vol}(\mathbb{B}(\mathbf{0}_d, 1))}{\int_{\mathbb{B}(\mathbf{0}_d,1)} \exp(-f(\mathbf{x}))\,\mathrm{d}\mathbf{x}}.$$

Since $f(\mathbf{x}) \leq f(\mathbf{0}) + 1$ for all $\mathbf{x} \in \mathbb{B}(\mathbf{0}_d, 1)$, we have $\exp(-f(\mathbf{x})) \geq \exp(-f(\mathbf{0}) - 1)$, so:

$$\int_{\mathbb{B}(\mathbf{0}_d,1)} \exp(-f(\mathbf{x}))\,\mathrm{d}\mathbf{x} \geq \exp(-f(\mathbf{0}) - 1) \cdot \mathrm{Vol}(\mathbb{B}(\mathbf{0}_d, 1)).$$

Therefore:

$$N \leq \frac{e \cdot \exp(-f(\mathbf{0})) \cdot \mathrm{Vol}(\mathbb{B}(\mathbf{0}_d, 1))}{\exp(-f(\mathbf{0}) - 1) \cdot \mathrm{Vol}(\mathbb{B}(\mathbf{0}_d, 1))} = \frac{e \cdot \exp(-f(\mathbf{0}))}{\exp(-f(\mathbf{0}) - 1)} = e \cdot e = e^2.$$

Including the initial query to compute $f(\mathbf{0})$, the total expected number of value oracle queries to f is $1 + e^2 = O(1)$.

Problem 3

(i)

The kernel of $\mathbf{L}_G$ is the span of the indicator vectors of the connected components of G.

Proof. By Lemma 10 (Part II),

$$\mathbf{x}^\top \mathbf{L}_G \mathbf{x} = \sum_{e=(u,v)\in E} w_e (x_u - x_v)^2.$$

Since $w_e > 0$ for all $e \in E$, this equals zero if and only if $x_u = x_v$ for every edge $(u, v) \in E$. Equivalently, $\mathbf{x}$ is constant on each connected component of G. The set of all such vectors is precisely $\mathrm{span}\{\mathbf{1}_{C_1}, \ldots, \mathbf{1}_{C_k}\}$, where $C_1, \ldots, C_k$ are the connected components of G. $\square$

(ii)

For $\mathbf{x} \in \{-\frac{1}{2}, \frac{1}{2}\}^V$, the vector $\mathbf{x}$ defines a cut $(S, V \setminus S)$ where $S = \{v \in V : x_v = \frac{1}{2}\}$. For each edge $e = (u, v)$, we have $(x_u - x_v)^2 = 1$ if e crosses the cut (i.e., exactly one of u, v is in S) and $(x_u - x_v)^2 = 0$ otherwise. By Lemma 10 (Part II),

$$\mathbf{x}^\top \mathbf{L}_G \mathbf{x} = \sum_{e=(u,v)\in E} w_e (x_u - x_v)^2 = \sum_{\substack{e=(u,v)\in E \\ |\{u,v\}\cap S|=1}} w_e.$$

This is the total weight of edges crossing the cut $(S, V \setminus S)$.

(iii)

Proof. Write the Laplacian as a sum of rank-one terms:

$$\mathbf{L}_G = \sum_{e=(u,v)\in E} w_e(\mathbf{e}_u - \mathbf{e}_v)(\mathbf{e}_u - \mathbf{e}_v)^\top.$$

Define the matrix $\mathbf{A} \in \mathbb{R}^{|E|\times|V|}$ whose row indexed by edge $e = (u, v)$ is $\mathbf{a}_e := \sqrt{w_e}(\mathbf{e}_u - \mathbf{e}_v)^\top$. Then

$$\mathbf{A}^\top\mathbf{A} = \sum_{e\in E} \mathbf{a}_e^\top \mathbf{a}_e = \sum_{e=(u,v)\in E} w_e(\mathbf{e}_u - \mathbf{e}_v)(\mathbf{e}_u - \mathbf{e}_v)^\top = \mathbf{L}_G.$$

We apply Proposition 3 (Part IX) to $\mathbf{A}$. Set $\varepsilon = 0.1$, $\delta = \frac{1}{2}$, $n = |E|$, and $d = |V|$.

Rank bound: $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{L}_G) \le |V|$.

Sample complexity: By Proposition 3 (Part IX), taking

$$K \ge \frac{3\operatorname{rank}(\mathbf{A})}{\varepsilon^2}\log\left(\frac{2d}{\delta}\right) \le \frac{3|V|}{0.01}\log(4|V|) = O(|V|\log|V|)$$

i.i.d. samples $e_1, \ldots, e_K$ from the leverage score distribution $p_e = \tau_e(\mathbf{A})/\|\boldsymbol{\tau}\|_1$ guarantees, with probability at least $\frac{1}{2}$, that

$$(1-\varepsilon)\,\mathbf{A}^\top\mathbf{A} \preceq \mathbf{P} \preceq (1+\varepsilon)\,\mathbf{A}^\top\mathbf{A},$$

where

$$\mathbf{P} = \sum_{k=1}^{K} \frac{1}{K\,p_{e_k}}\mathbf{a}_{e_k}^\top \mathbf{a}_{e_k} = \sum_{k=1}^{K} \frac{w_{e_k}}{K\,p_{e_k}}(\mathbf{e}_{u_k} - \mathbf{e}_{v_k})(\mathbf{e}_{u_k} - \mathbf{e}_{v_k})^\top.$$

Identifying the sparsifier: Each term in $\mathbf{P}$ has the form $w'(\mathbf{e}_u - \mathbf{e}_v)(\mathbf{e}_u - \mathbf{e}_v)^\top$ for some edge $(u, v) \in E$ with positive weight $w' = w_{e_k}/(K\,p_{e_k}) > 0$. If the same edge is sampled multiple times, we sum its weights. Hence $\mathbf{P} = \mathbf{L}_H$ for a reweighted subgraph $H = (V, E', \mathbf{w}')$ with $E' \subseteq E$ and $\mathbf{w}' \in \mathbb{R}^{E'}_{>0}$. The number of distinct edges satisfies $|E'| \le K = O(|V|\log|V|)$.

Since $\mathbf{A}^\top\mathbf{A} = \mathbf{L}_G$ and $\varepsilon = 0.1$, the guarantee becomes

$$0.9\,\mathbf{L}_G \preceq \mathbf{L}_H \preceq 1.1\,\mathbf{L}_G.$$

The sampling succeeds with probability at least $\frac{1}{2} > 0$, so such a graph H exists. □

Problem 4

Set $\pi_0 = \mathcal{N}\big(\mathbf{x}^\star, \frac{1}{L}\mathbf{I}_d\big)$, the Gaussian centered at the minimizer with covariance $\frac{1}{L}\mathbf{I}_d$. We show π_0 is β-warm with respect to $\pi^\star$ for $\beta = \kappa^{d/2} = \exp\big(\frac{d}{2}\log\kappa\big)$.

Proof. Write the normalizing constant $Z := \int_{\mathbb{R}^d} \exp(-f(\mathbf{x}))\,d\mathbf{x}$, so that

$$\pi^\star(\mathbf{x}) = \frac{\exp(-f(\mathbf{x}))}{Z}.$$

The Gaussian density is

$$\pi_0(\mathbf{x}) = \left(\frac{L}{2\pi}\right)^{d/2} \exp\left(-\frac{L}{2}\|\mathbf{x}-\mathbf{x}^\star\|_2^2\right).$$

The warmness ratio at any $\mathbf{x} \in \mathbb{R}^d$ is

$$\frac{\pi_0(\mathbf{x})}{\pi^\star(\mathbf{x})} = \left(\frac{L}{2\pi}\right)^{d/2} \exp\left(-\frac{L}{2}\|\mathbf{x}-\mathbf{x}^\star\|_2^2 + f(\mathbf{x})\right) \cdot Z.$$

Bounding the exponent. Since f is L-smooth and $\mathbf{x}^\star$ is the minimizer (so $\nabla f(\mathbf{x}^\star) = \mathbf{0}$), the quadratic upper bound gives

$$f(\mathbf{x}) \le f(\mathbf{x}^\star) + \frac{L}{2}\|\mathbf{x}-\mathbf{x}^\star\|_2^2.$$

Therefore, for every $\mathbf{x} \in \mathbb{R}^d$,

$$-\frac{L}{2}\|\mathbf{x}-\mathbf{x}^\star\|_2^2 + f(\mathbf{x}) \le f(\mathbf{x}^\star).$$

Bounding Z. Since f is μ-strongly convex with minimizer $\mathbf{x}^\star$ (so $\nabla f(\mathbf{x}^\star) = \mathbf{0}$),

$$f(\mathbf{x}) \ge f(\mathbf{x}^\star) + \frac{\mu}{2}\|\mathbf{x}-\mathbf{x}^\star\|_2^2 \quad \text{for all } \mathbf{x} \in \mathbb{R}^d.$$

Exponentiating and integrating:

$$\begin{aligned}
Z &= \int_{\mathbb{R}^d} \exp(-f(\mathbf{x}))\, d\mathbf{x} \\
&\le \int_{\mathbb{R}^d} \exp\left(-f(\mathbf{x}^\star) - \frac{\mu}{2}\|\mathbf{x}-\mathbf{x}^\star\|_2^2\right) d\mathbf{x} \\
&= \exp(-f(\mathbf{x}^\star)) \int_{\mathbb{R}^d} \exp\left(-\frac{\mu}{2}\|\mathbf{x}-\mathbf{x}^\star\|_2^2\right) d\mathbf{x} \\
&= \exp(-f(\mathbf{x}^\star)) \left(\frac{2\pi}{\mu}\right)^{d/2}.
\end{aligned}$$

Combining. Substituting both bounds into the ratio:

$$\begin{aligned}
\frac{\pi_0(\mathbf{x})}{\pi^\star(\mathbf{x})} &= \left(\frac{L}{2\pi}\right)^{d/2} \exp\left(-\frac{L}{2}\|\mathbf{x}-\mathbf{x}^\star\|_2^2 + f(\mathbf{x})\right) \cdot Z \\
&\le \left(\frac{L}{2\pi}\right)^{d/2} \exp(f(\mathbf{x}^\star)) \cdot \exp(-f(\mathbf{x}^\star)) \left(\frac{2\pi}{\mu}\right)^{d/2} \\
&= \left(\frac{L}{2\pi}\right)^{d/2} \cdot \left(\frac{2\pi}{\mu}\right)^{d/2} \\
&= \left(\frac{L}{\mu}\right)^{d/2} = \kappa^{d/2}.
\end{aligned}$$

This holds for every $\mathbf{x} \in \mathbb{R}^d$, so by Definition 2 (Part XV), π_0 is β-warm with respect to $\pi^\star$ with

$$\beta = \kappa^{d/2} = \exp\left(\frac{d}{2}\log\kappa\right) = \exp(O(d\log\kappa)).$$

□

Problem 5

Proof. Set $\pi_k := T^k \pi_0$ and $\delta_k := D_{\mathrm{TV}}(\pi_k, \pi^\star)$.

By Lemma 2 (Part XV), since π_0 is β-warm with respect to $\pi^\star$, π_k is β-warm with respect to $\pi^\star$ for all $k \geq 0$ (apply the lemma inductively: $T\pi_k$ is β-warm whenever π_k is).

Claim: If $\delta_k \geq \epsilon$, then $\delta_{k+1} \leq \delta_k/2$.

Since $D_{\mathrm{TV}}(\pi_k, \pi^\star) = \delta_k$, decompose via the TV coupling:

$$\pi_k = (1-\delta_k)\sigma_k + \delta_k \rho_k, \qquad \pi^\star = (1-\delta_k)\sigma_k + \delta_k \rho_k',$$

where

$$\sigma_k(x) = \frac{\min(\pi_k(x), \pi^\star(x))}{1-\delta_k}, \quad \rho_k(x) = \frac{(\pi_k(x) - \pi^\star(x))^+}{\delta_k}, \quad \rho_k'(x) = \frac{(\pi^\star(x) - \pi_k(x))^+}{\delta_k}.$$

These are valid densities: $\sigma_k, \rho_k, \rho_k' \geq 0$ by construction, and integrating the decomposition of π_k gives

$$1 = (1-\delta_k)\int \sigma_k + \delta_k \int \rho_k,$$

while $\int \sigma_k = \frac{\int \min(\pi_k, \pi^\star)}{1-\delta_k} = \frac{1-\delta_k}{1-\delta_k} = 1$ (since $\int \min(\pi_k, \pi^\star) = 1 - D_{\mathrm{TV}}(\pi_k, \pi^\star) = 1 - \delta_k$), so $\int \rho_k = 1$. The same argument gives $\int \rho_k' = 1$.

By linearity of the transition operator T and stationarity $T\pi^\star = \pi^\star$:

$$\begin{aligned} T\pi_k &= (1-\delta_k)T\sigma_k + \delta_k T\rho_k, \\ \pi^\star = T\pi^\star &= (1-\delta_k)T\sigma_k + \delta_k T\rho_k'. \end{aligned}$$

Subtracting, for any measurable S:

$$T\pi_k(S) - \pi^\star(S) = \delta_k\big(T\rho_k(S) - T\rho_k'(S)\big).$$

Taking $\sup_S$ on both sides:

$$\delta_{k+1} = D_{\mathrm{TV}}(T\pi_k, \pi^\star) = \delta_k \cdot D_{\mathrm{TV}}(T\rho_k, T\rho_k').$$

For ρ_k:

$$\rho_k(x) = \frac{(\pi_k(x) - \pi^\star(x))^+}{\delta_k} \leq \frac{\pi_k(x)}{\delta_k} \leq \frac{\beta\pi^\star(x)}{\delta_k},$$

so ρ_k is $\frac{\beta}{\delta_k}$-warm with respect to $\pi^\star$. Since $\delta_k \geq \epsilon$, ρ_k is $\frac{\beta}{\epsilon}$-warm. For ρ_k':

$$\rho_k'(x) = \frac{(\pi^\star(x) - \pi_k(x))^+}{\delta_k} \leq \frac{\pi^\star(x)}{\delta_k},$$

so ρ_k' is $\frac{1}{\delta_k}$-warm, hence $\frac{\beta}{\epsilon}$-warm since $\beta \geq 1$ and $\delta_k \geq \epsilon$.

By the hypothesis on the random walk, any $\frac{\beta}{\epsilon}$-warm density π satisfies $D_{\mathrm{TV}}(T\pi, \pi^\star) \leq \frac{1}{4}$. Applying this to ρ_k and ρ_k':

$$D_{\mathrm{TV}}(T\rho_k, \pi^\star) \leq \frac{1}{4}, \qquad D_{\mathrm{TV}}(T\rho_k', \pi^\star) \leq \frac{1}{4}.$$

By the triangle inequality:

$$D_{\mathrm{TV}}(T\rho_k, T\rho'_k) \le D_{\mathrm{TV}}(T\rho_k, \pi^\star) + D_{\mathrm{TV}}(\pi^\star, T\rho'_k) \le \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Therefore $\delta_{k+1} = \delta_k \cdot D_{\mathrm{TV}}(T\rho_k, T\rho'_k) \le \delta_k/2$, proving the claim.

We show by induction that $\delta_k \le 2^{-(k+1)}$ for all $k \ge 1$. The base case $k = 1$: since π_0 is β-warm and $\epsilon < 1$ implies $\frac{\beta}{\epsilon} \ge \beta \ge 1$, π_0 is $\frac{\beta}{\epsilon}$-warm, so the hypothesis gives $\delta_1 \le \frac{1}{4} = 2^{-2}$. For the inductive step, suppose $\delta_k \le 2^{-(k+1)}$. If $\delta_k < \epsilon$, then $\delta_{k+1} \le \delta_k < \epsilon$ by Lemma 2 (Part XV) (TV distance does not increase under T), so the conclusion holds. If $\delta_k \ge \epsilon$, the halving gives $\delta_{k+1} \le \delta_k/2 \le 2^{-(k+2)}$.

For $k \ge \log_2(1/\epsilon)$:

$$\delta_k \le 2^{-(k+1)} \le 2^{-(\log_2(1/\epsilon)+1)} = \frac{\epsilon}{2} \le \epsilon.$$

Hence $D_{\mathrm{TV}}(T^k\pi_0, \pi^\star) \le \epsilon$ whenever $k \ge \log_2(1/\epsilon)$. □